

ClassicModels MySQL Metadata for RAG-Based Natural-Language-to-SQL

Canonical metadata for retrieval, context selection, and efficient SQL generation

Field	Value
Metadata Version	1.0.0
Generated On	2026-07-14
Canonical Database	classicmodels
Source Schema Version	3.2
Intended Sql Dialect	MySQL
Intended Generation Model	Gemini 3.1 Flash-Lite
Scope	Read/query metadata only. It does not contain application code or executable SQL examples.

The JSON, TXT, DOCX, and PDF deliverables are generated from this same canonical metadata object.

1. Purpose and Usage

Provide complete schema, semantic, relationship, metric, query-planning, and retrieval metadata so a RAG pipeline can retrieve the right context before generating SQL.

This document is designed to be ingested as metadata context for a RAG pipeline. It combines exact schema facts with business semantics, user-language synonyms, calculation rules, join guidance, and known limitations. It intentionally contains no executable SQL examples.

2. Database Overview

Property	Value
Business Domain	Fictional retailer of scale models of classic cars and other vehicles.
Storage Engine	InnoDB
Character Set	utf8
Collation	utf8_general_ci
Table Count	8
Column Count	59
Standard Total Row Count	3864
Historical Time Span	Order and payment activity is historical sample data mainly covering 2003 through 2005.
Currency Note	No currency-code column exists. Monetary columns are normally treated as one common currency in the sample database, but the SQL generator must not label the currency unless the application defines it.

Business Processes Covered

- Product catalog and product-line classification
- Inventory snapshot
- Customers and assigned sales representatives
- Sales orders and order line items
- Customer payments
- Employees, reporting hierarchy, and sales offices

Standard Row Counts

Table	Rows
productlines	7
products	110
employees	23
offices	7
customers	122
orders	326
orderdetails	2996

Table	Rows
payments	273

Major Absences

- No order-level total column
- No tax, shipping fee, return, refund, or coupon table
- No direct payment-to-order allocation
- No inventory movement history or historical stock snapshots
- No supplier master table beyond productVendor text
- No customer territory column
- No created-at or updated-at timestamps

Identifier Conventions

- Use exact lower-case table names from the schema.
- Use exact camelCase column names from the schema.
- Identifiers such as productCode, officeCode, postalCode, phone, extension, and checkNumber are text, even when they contain digits.
- No table declares AUTO_INCREMENT in the source schema.

3. Semantic Entity Map

Entity	Primary Table	Identifier	Display / Relationship
Customer	customers	customerNumber	customerName
Customer contact	customers	customerNumber	contactFirstName + contactLastName
Sales representative	employees	employeeNumber	firstName + lastName
Manager	employees	employeeNumber	firstName + lastName employees.reportsTo references employees.employeeNumber
Office	offices	officeCode	city
Order	orders	orderNumber	orderNumber
Order line	orderdetails	orderNumber + productCode	productCode within orderNumber
Product	products	productCode	productName
Product line	productlines	productLine	productLine
Payment	payments	customerNumber + checkNumber	checkNumber

4. Complete Table and Column Metadata

4.1 offices

Business aliases: office; branch; sales office; location; regional office

Purpose: Stores sales-office location and territory information.

Row grain: One row per sales office.

Standard row count: 7

Primary key: officeCode

Foreign keys:

Indexed columns: PRIMARY KEY (officeCode)

Common query intents: office lookup; office geography; employees by office; office performance; territory analysis

Common filters: officeCode; city; country; territory

Common groupings: city; country; territory

Table Notes

- Office performance requires joining offices to employees, then usually customers, orders, and orderdetails.
- Territory is an office attribute, not a customer attribute.
- officeCode is text even though standard values look numeric.

Columns

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
officeCode varchar(10)	Nullable: False Primary key Role: identifier	Unique office identifier.	Synonyms: office id; branch code; office number Values: 1; 4; 7	Aggregation: Count distinct for office counts; do not sum. Filter: Use exact matching unless the user provides only a partial code. Null: Not applicable; column is required. Join: Join to employees.officeCode.
city varchar(50)	Nullable: False Non-key attribute Role: geography	City where the office is located.	Synonyms: office city; branch city; location city Values: San Francisco; Paris; Tokyo	Aggregation: Group or count; do not sum. Filter: Use case-insensitive text matching and normalize common city abbreviations only when unambiguous. Null: Not applicable; column is required. Join: Not a join key.
phone varchar(50)	Nullable: False Non-key attribute Role: contact detail	Office telephone number.	Synonyms: office phone; branch phone; telephone Values: +1 650 219 4782; +33 14 723 4404	Aggregation: Do not aggregate. Filter: Use exact or contains matching; preserve punctuation and country codes. Null: Not applicable; column is required. Join: Not a join key. Cautions: Text identifier; never cast to a number for arithmetic.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
addressLine1 varchar(50)	Nullable: False Non-key attribute Role: address	Primary street-address line for the office.	Synonyms: street address; office address; address line 1 Values: 100 Market Street; 43 Rue Jouffroy D'abbans	Aggregation: Do not aggregate. Filter: Use text search for address lookup. Null: Not applicable; column is required. Join: Not a join key.
addressLine2 varchar(50)	Nullable: True Non-key attribute Role: address	Optional secondary office-address line such as suite or floor.	Synonyms: suite; floor; address line 2 Values: Suite 300; Floor #2	Aggregation: Do not aggregate. Filter: Use IS NULL semantics for offices without a second address line. Null: No secondary address line is recorded. Join: Not a join key.
state varchar(50)	Nullable: True Non-key attribute Role: geography	State, province, prefecture, or comparable administrative area.	Synonyms: state; province; region Values: CA; MA; Chiyoda-Ku	Aggregation: Group or count; do not sum. Filter: Do not assume NULL means missing data; many countries do not use this field in the sample. Null: Administrative subdivision is not recorded or not applicable. Join: Not a join key.
country varchar(50)	Nullable: False Non-key attribute Role: geography	Country of the office.	Synonyms: office country; nation Values: USA; France; Japan; Australia; UK	Aggregation: Group, count, or count distinct. Filter: Use exact normalized country values from the data; do not silently merge UK and USA-style abbreviations. Null: Not applicable; column is required. Join: Not a join key.
postalCode varchar(15)	Nullable: False Non-key attribute Role: geography identifier	Office postal or ZIP code.	Synonyms: postal code; zip code; postcode Values: 94080; 02107; EC2N 1HN	Aggregation: Do not aggregate. Filter: Treat as text and preserve leading zeros and spaces. Null: Not applicable; column is required. Join: Not a join key. Cautions: Never perform numeric ranges unless explicitly requested and semantically valid.
territory varchar(10)	Nullable: False Non-key attribute Role: organizational geography	Office sales territory label.	Synonyms: territory; sales region; business region Values: NA; EMEA; Japan; APAC	Aggregation: Group or count. Filter: Use exact values. Note that Japan is stored as its own territory label rather than APAC. Null: Not applicable; column is required. Join: Not a join key.

4.2 employees

Business aliases: employee; staff; sales rep; representative; manager; executive

Purpose: Stores employees, office assignments, job titles, and the management reporting hierarchy.

Row grain: One row per employee.

Standard row count: 23

Primary key: employeeNumber

Foreign keys: {'columns': ['officeCode'], 'references': 'offices.officeCode', 'optional': False}; {'columns': ['reportsTo'], 'references': 'employees.employeeNumber', 'optional': True, 'relationship_type': 'self-reference'}

Indexed columns: PRIMARY KEY (employeeNumber); INDEX (reportsTo); INDEX (officeCode)

Common query intents: employee lookup; sales representative lookup; manager hierarchy; direct reports; employees by office; employee sales performance

Common filters: employeeNumber; firstName; lastName; jobTitle; officeCode; reportsTo

Common groupings: jobTitle; officeCode; reportsTo

Table Notes

- The employee hierarchy is represented by reportsTo and may require repeated self-joins or recursive logic for multiple levels.
- Only employees referenced by customers.salesRepEmployeeNumber are customer-facing assigned representatives.
- Employee performance is not stored directly; it is derived through assigned customers and their orders.

Columns

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
employeeNumber int(11)	Nullable: False Primary key Role: identifier	Unique employee identifier.	Synonyms: employee id; staff id; rep number; sales rep id Values: 1002; 1370; 1702	Aggregation: Count distinct for employee counts; do not sum. Filter: Use exact numeric matching. Null: Not applicable; column is required. Join: Join from customers.salesRepEmployeeNumber or employees.reportsTo.
lastName varchar(50)	Nullable: False Non-key attribute Role: person name	Employee family name.	Synonyms: surname; family name; employee last name Values: Murphy; Patterson; Hernandez	Aggregation: Do not aggregate. Filter: Use case-insensitive text matching; combine with firstName when names may be ambiguous. Null: Not applicable; column is required. Join: Not a join key.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
firstName varchar(50)	Nullable: False Non-key attribute Role: person name	Employee given name.	Synonyms: given name; employee first name Values: Diane; Mary; Gerard	Aggregation: Do not aggregate. Filter: Use case-insensitive text matching; combine with lastName for person lookup. Null: Not applicable; column is required. Join: Not a join key.
extension varchar(10)	Nullable: False Non-key attribute Role: contact detail	Internal telephone extension.	Synonyms: extension; phone extension; ext Values: x5800; x2028	Aggregation: Do not aggregate. Filter: Treat as text; preserve the leading x. Null: Not applicable; column is required. Join: Not a join key. Cautions: Not a numeric measure.
email varchar(100)	Nullable: False Non-key attribute Role: contact detail	Employee email address.	Synonyms: email; employee email; mail address Values: dmurphy@classicmodelcars.com	Aggregation: Do not aggregate. Filter: Use exact or case-insensitive matching. Null: Not applicable; column is required. Join: Not a join key.
officeCode varchar(10)	Nullable: False Foreign key Role: relationship key	Office to which the employee is assigned.	Synonyms: office id; branch code; employee office Values: 1; 4; 7	Aggregation: Group or count by office; do not sum. Filter: Filter by exact officeCode or join to offices for city/country/territory. Null: Not applicable; column is required. Join: Join to offices.officeCode.
reportsTo int(11)	Nullable: True Self-referencing foreign key Role: hierarchy key	Employee number of the employee's manager.	Synonyms: manager id; supervisor id; reports to; boss Values: 1002; 1056; 1102	Aggregation: Do not sum. Count direct reports by grouping on this column. Filter: Use NULL to identify the top-level employee. Null: Employee has no recorded manager; in the standard data this identifies the President. Join: Self-join to employees.employeeNumber to retrieve manager details.
jobTitle varchar(50)	Nullable: False Non-key attribute Role: organizational category	Employee role or job title.	Synonyms: role; designation; position; job Values: President; VP Sales; VP Marketing; Sales Manager (APAC); Sale Manager (EMEA); Sales Manager (NA); Sales Rep	Aggregation: Group or count. Filter: Use exact stored titles when possible. Preserve the source-data spelling "Sale Manager (EMEA)". Null: Not applicable; column is required. Join: Not a join key.

4.3 customers

Business aliases: customer; client; account; buyer; company

Purpose: Stores customer organizations, contact information, geography, assigned sales representative, and credit limit.

Row grain: One row per customer organization.

Standard row count: 122

Primary key: customerNumber

Foreign keys: {'columns': ['salesRepEmployeeNumber'], 'references': 'employees.employeeNumber', 'optional': True}

Indexed columns: PRIMARY KEY (customerNumber); INDEX (salesRepEmployeeNumber)

Common query intents: customer lookup; customer geography; customer credit; assigned representative; customers with or without orders; customer value; customer payment history

Common filters: customerNumber; customerName; contact names; city; state; country; salesRepEmployeeNumber; creditLimit

Common groupings: country; city; salesRepEmployeeNumber; creditLimit bands

Table Notes

- Customer organization and customer contact are represented in the same row.
- Customer sales and payments require separate joins; avoid multiplying totals by joining orders and payments at detail level in one aggregation.
- Some customers are unassigned to a sales representative and can be lost by inner joins.

Columns

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
customerNumber int(11)	Nullable: False Primary key Role: identifier	Unique customer identifier.	Synonyms: customer id; client id; account number Values: 103; 141; 496	Aggregation: Count distinct for customer counts; do not sum. Filter: Use exact numeric matching. Null: Not applicable; column is required. Join: Join to orders.customerNumber and payments.customerNumber.
customerName varchar(50)	Nullable: False Non-key attribute Role: entity name	Customer company or organization name.	Synonyms: customer name; client name; company name; account name Values: Atelier graphique; Euro+ Shopping Channel; Kelly's Gift Shop	Aggregation: Do not aggregate. Filter: Use case-insensitive exact or contains matching; prefer exact match when a full name is provided. Null: Not applicable; column is required. Join: Not a join key.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
contactLastName varchar(50)	Nullable: False Non-key attribute Role: person name	Family name of the customer's primary contact person.	Synonyms: contact surname; contact last name Values: Schmitt; Freyre; Snowden	Aggregation: Do not aggregate. Filter: Combine with contactFirstName for contact-person lookup. Null: Not applicable; column is required. Join: Not a join key.
contactFirstName varchar(50)	Nullable: False Non-key attribute Role: person name	Given name of the customer's primary contact person.	Synonyms: contact first name; contact given name Values: Carine; Diego; Tony	Aggregation: Do not aggregate. Filter: Trim whitespace when displaying or comparing because some sample values contain trailing spaces. Null: Not applicable; column is required. Join: Not a join key.
phone varchar(50)	Nullable: False Non-key attribute Role: contact detail	Customer telephone number.	Synonyms: customer phone; client telephone; contact number Values: 40.32.2555; 7025551838; +64 9 5555500	Aggregation: Do not aggregate. Filter: Treat as text; punctuation and country-code formats vary. Null: Not applicable; column is required. Join: Not a join key.
addressLine1 varchar(50)	Nullable: False Non-key attribute Role: address	Primary customer street-address line.	Synonyms: customer address; street address; address line 1 Values: 54, rue Royale; 636 St Kilda Road	Aggregation: Do not aggregate. Filter: Use text matching only when needed. Null: Not applicable; column is required. Join: Not a join key.
addressLine2 varchar(50)	Nullable: True Non-key attribute Role: address	Optional secondary address line.	Synonyms: suite; floor; address line 2 Values: Level 3; Suite 400	Aggregation: Do not aggregate. Filter: Use IS NULL semantics to find customers without a second line. Null: No secondary address line is recorded. Join: Not a join key.
city varchar(50)	Nullable: False Non-key attribute Role: geography	Customer city.	Synonyms: customer city; client city Values: Nantes; Las Vegas; Singapore	Aggregation: Group, count, or count distinct. Filter: Use case-insensitive text matching; normalize abbreviations only when unambiguous. Null: Not applicable; column is required. Join: Not a join key.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
state varchar(50)	Nullable: True Non-key attribute Role: geography	Customer state, province, or comparable subdivision.	Synonyms: state; province; region Values: NV; Victoria; Osaka	Aggregation: Group or count. Filter: NULL is common for countries where this field is not used in the sample. Null: Subdivision is not recorded or not applicable. Join: Not a join key.
postalCode varchar(15)	Nullable: True Non-key attribute Role: geography identifier	Customer postal or ZIP code.	Synonyms: postal code; zip code; postcode Values: 44000; 83030; EC2 5NT	Aggregation: Do not aggregate. Filter: Treat as text and preserve leading zeros, spaces, and prefixes. Null: Postal code is not recorded. Join: Not a join key. Cautions: Never cast to numeric for ordinary filtering.
country varchar(50)	Nullable: False Non-key attribute Role: geography	Customer country.	Synonyms: customer country; client nation; market country Values: France; USA; Australia; Spain	Aggregation: Group, count, or count distinct. Filter: Use exact stored values; trim unexpected trailing spaces when comparing or displaying. Null: Not applicable; column is required. Join: Not a join key.
salesRepEmployeeNumber int(11)	Nullable: True Foreign key Role: relationship key	Employee assigned as the customer's sales representative.	Synonyms: sales rep; account manager; representative id; employee assigned Values: 1165; 1370; 1612	Aggregation: Count customers per representative; do not sum. Filter: Use NULL to identify unassigned customers. Null: Customer has no assigned sales representative. Join: Join to employees.employeeNumber.
creditLimit decimal(10,2)	Nullable: True Non-key attribute Role: monetary measure	Customer's approved credit limit.	Synonyms: credit limit; credit ceiling; approved credit Values: 21000.00; 227600.00; 0.00	Aggregation: May be summed or averaged only when the business question asks about total or average approved credit; otherwise display per customer. Filter: Use numeric comparisons. Distinguish zero from NULL. Null: Credit limit is not recorded. Join: Not a join key. Cautions: Credit limit is not customer balance, revenue, payment total, or available credit.

4.4 payments

Business aliases: payment; receipt; cash received; check; customer payment

Purpose: Stores payments received from customers.

Row grain: One row per customer payment/check transaction.

Standard row count: 273

Primary key: customerNumber; checkNumber

Foreign keys: {'columns': ['customerNumber'], 'references': 'customers.customerNumber', 'optional': False}

Indexed columns: PRIMARY KEY (customerNumber, checkNumber)

Common query intents: payment history; payments by customer; payments by period; total paid; average payment; customer payment behavior

Common filters: customerNumber; checkNumber; paymentDate; amount

Common groupings: customerNumber; payment year/month; amount bands

Table Notes

- There is no orderNumber in payments, so direct payment-to-order matching is impossible from this schema alone.
- When combining customer order totals and payment totals, aggregate each side separately by customer before joining.
- A payment may cover multiple orders or account-level balances; do not invent allocation logic.

Columns

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
customerNumber int(11)	Nullable: False Primary-key component and foreign key Role: identifier and relationship key	Customer who made the payment.	Synonyms: customer id; payer; account number Values: 103; 141; 496	Aggregation: Group payment measures by customer; do not sum the identifier. Filter: Use exact customer number or join to customers for names. Null: Not applicable; column is required. Join: Join to customers.customerNumber.
checkNumber varchar(50)	Nullable: False Primary-key component Role: transaction identifier	Payment/check reference identifier.	Synonyms: check number; payment reference; receipt number; transaction reference Values: HQ336336; JM555205; EU531600	Aggregation: Count or count distinct; do not sum. Filter: Use exact text matching. Null: Not applicable; column is required. Join: Not a join key. Cautions: Not guaranteed to be globally unique without customerNumber because the primary key is composite.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
paymentDate date	Nullable: False Non-key attribute Role: date	Date the payment was received or recorded.	Synonyms: payment date; receipt date; paid on Values: 2003-06-05; 2004-10-19; 2005-05-25	Aggregation: Group by year, quarter, month, or date. Filter: Use this date for cash-receipt timing, not orderDate or shippedDate. Null: Not applicable; column is required. Join: Not a join key.
amount decimal(10,2)	Nullable: False Non-key attribute Role: monetary additive measure	Amount of the customer payment.	Synonyms: payment amount; amount paid; cash received; receipt value Values: 6066.78; 14571.44; 52166.00	Aggregation: Additive across payment rows. Sum for total payments; average for average payment size. Filter: Use numeric comparisons and aggregate after selecting the correct customer/date scope. Null: Not applicable; column is required. Join: Not a join key. Cautions: Currency code is not stored.; Payments are not linked to individual orders.

4.5 productlines

Business aliases: product line; category; product category; vehicle category

Purpose: Stores the seven product categories and their descriptive content.

Row grain: One row per product line/category.

Standard row count: 7

Primary key: productLine

Foreign keys:

Indexed columns: PRIMARY KEY (productLine)

Common query intents: product-line lookup; category descriptions; products by category; sales by product line

Common filters: productLine; textDescription

Common groupings: productLine

Table Notes

- The exact seven categories are Classic Cars, Vintage Cars, Motorcycles, Trucks and Buses, Planes, Ships, and Trains.
- Use products for product-level price, vendor, scale, and inventory analysis.

Columns

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
productLine varchar(50)	Nullable: False Primary key Role: category identifier	Name of the product category.	Synonyms: product line; category; vehicle type Values: Classic Cars; Vintage Cars; Motorcycles; Trucks and Buses; Planes; Ships; Trains	Aggregation: Group or count. Do not sum. Filter: Use exact category values or safe case-insensitive matching. Null: Not applicable; column is required. Join: Join to products.productLine.
textDescription varchar(4000)	Nullable: True Non-key attribute Role: unstructured descriptive text	Plain-text description of the product line.	Synonyms: category description; product-line text; description Values: Long descriptive paragraph	Aggregation: Do not aggregate. Filter: Use full-text or contains matching only for semantic description questions. Null: No plain-text description is stored. Join: Not a join key.
htmlDescription mediumtext	Nullable: True Non-key attribute Role: unstructured HTML text	Optional HTML-formatted product-line description.	Synonyms: HTML description; web description Values: NULL in the standard dataset	Aggregation: Do not aggregate. Filter: Normally exclude from analytics unless the user explicitly asks for web/HTML content. Null: No HTML description is stored in the standard dataset. Join: Not a join key. Cautions: May contain markup rather than clean business text.
image mediumblob	Nullable: True Non-key attribute Role: binary content	Optional binary image for the product line.	Synonyms: image; picture; category image Values: NULL in the standard dataset	Aggregation: Do not aggregate. Filter: Do not select by default; retrieve only when binary content is explicitly requested. Null: No image is stored in the standard dataset. Join: Not a join key. Cautions: Not suitable for ordinary text-to-SQL result display.

4.6 products

Business aliases: product; model; item; SKU; catalog item

Purpose: Stores the product catalog, product-line membership, model scale, vendor, description, stock quantity, cost, and list price.

Row grain: One row per product/model SKU.

Standard row count: 110

Primary key: productCode

Foreign keys: {'columns': ['productLine'], 'references': 'productlines.productLine', 'optional': False}

Indexed columns: PRIMARY KEY (productCode); INDEX (productLine)

Common query intents: product lookup; catalog search; inventory; pricing; margin potential; products by vendor; products by scale; sales by product

Common filters: productCode; productName; productLine; productScale; productVendor; quantityInStock; buyPrice; MSRP

Common groupings: productLine; productVendor; productScale; stock bands; price bands

Table Notes

- Actual quantity sold and sale price are in orderdetails.
- Potential unit margin can be estimated as MSRP minus buyPrice; realized unit margin uses priceEach minus buyPrice.
- A product with no orderdetails has no recorded sales in the sample period.

Columns

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
productCode varchar(15)	Nullable: False Primary key Role: identifier	Unique product/SKU code.	Synonyms: product id; SKU; item code; model code Values: S10_1678; S18_1749; S700_4002	Aggregation: Count distinct for product counts; do not sum. Filter: Use exact text matching; preserve underscores and prefixes. Null: Not applicable; column is required. Join: Join to orderdetails.productCode.
productName varchar(70)	Nullable: False Non-key attribute Role: entity name	Display name of the scale model product.	Synonyms: product name; model name; item name Values: 1968 Ford Mustang; The Titanic; 2003 Harley-Davidson Eagle Drag Bike	Aggregation: Do not aggregate. Filter: Use case-insensitive exact or contains matching; product names may include years and punctuation. Null: Not applicable; column is required. Join: Not a join key.
productLine varchar(50)	Nullable: False Foreign key Role: category relationship key	Product category to which the product belongs.	Synonyms: product line; category; vehicle type Values: Classic Cars; Ships; Planes	Aggregation: Group or count. Filter: Use exact known product-line values or join to productlines for descriptions. Null: Not applicable; column is required. Join: Join to productlines.productLine.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
productScale varchar(10)	Nullable: False Non-key attribute Role: product attribute	Scale ratio of the physical model.	Synonyms: scale; model scale; size ratio Values: 1:10; 1:18; 1:72; 1:700	Aggregation: Group or count; do not sum as text. Filter: Treat as a categorical string unless explicitly parsed into a denominator. Null: Not applicable; column is required. Join: Not a join key. Cautions: Lexicographic ordering is not the same as numeric scale ordering.
productVendor varchar(50)	Nullable: False Non-key attribute Role: organization name	Vendor or manufacturer name stored directly on the product.	Synonyms: vendor; manufacturer; supplier; brand Values: Welly Diecast Productions; Second Gear Diecast; Unimax Art Galleries	Aggregation: Group or count. Filter: Use case-insensitive text matching. Null: Not applicable; column is required. Join: Not a join key. Cautions: There is no separate vendor table or vendor identifier.
productDescription text	Nullable: False Non-key attribute Role: unstructured descriptive text	Long text describing product features and characteristics.	Synonyms: description; features; product details Values: Detailed product feature paragraph	Aggregation: Do not aggregate. Filter: Use contains/full-text semantics for feature searches; avoid returning the entire text unless requested. Null: Not applicable; column is required. Join: Not a join key.
quantityInStock smallint(6)	Nullable: False Non-key attribute Role: inventory measure	Current inventory quantity snapshot for the product.	Synonyms: stock; inventory; units in stock; available quantity Values: 68; 3252; 9997	Aggregation: Additive across different products for a current total-unit snapshot, but not across time because no history exists. Filter: Use numeric thresholds for low-stock or high-stock questions. Null: Not applicable; column is required. Join: Not a join key. Cautions: Not historical stock at order date.; No reserved or committed inventory field exists.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
buyPrice decimal(10,2)	Nullable: False Non-key attribute Role: monetary semi-additive measure	Unit purchase/cost price of the product.	Synonyms: cost; buying price; purchase price; unit cost Values: 31.92; 95.59; 103.42	Aggregation: Average or compare across products; multiply by quantities for estimated cost or inventory cost value. Filter: Use numeric comparisons. Null: Not applicable; column is required. Join: Not a join key. Cautions: Static product-level cost, not a historical cost captured on each order line.; Currency code is not stored.
MSRP decimal(10,2)	Nullable: False Non-key attribute Role: monetary semi-additive measure	Manufacturer's suggested retail/list price per unit.	Synonyms: MSRP; list price; retail price; suggested price Values: 79.80; 147.74; 207.80	Aggregation: Average or compare across products; multiply by quantities for list-price value. Filter: Use numeric comparisons and exact capitalization of the column name. Null: Not applicable; column is required. Join: Not a join key. Cautions: MSRP is not actual sales revenue; actual unit sale price is orderdetails.priceEach.

4.7 orders

Business aliases: order; sales order; purchase; order header

Purpose: Stores sales-order header information including dates, status, comments, and customer.

Row grain: One row per sales order.

Standard row count: 326

Primary key: orderNumber

Foreign keys: {'columns': ['customerNumber'], 'references': 'customers.customerNumber', 'optional': False}

Indexed columns: PRIMARY KEY (orderNumber); INDEX (customerNumber)

Common query intents: order lookup; order status; orders by customer; orders by date; shipping performance; open/unshipped orders; delayed orders

Common filters: orderNumber; customerNumber; orderDate; requiredDate; shippedDate; status; comments

Common groupings: customerNumber; status; order year/month; shipment year/month

Table Notes

- Order value is not stored in this table; derive it from orderdetails.
- Status applies to the whole order, not individual line items.

- Cancelled or disputed orders may still have orderdetail rows; include or exclude statuses according to the business question.
- Order dates run from 2003-01-06 through 2005-05-31 in the standard dataset.

Columns

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
orderNumber int(11)	Nullable: False Primary key Role: identifier	Unique sales-order identifier.	Synonyms: order id; sales order number; order number Values: 10100; 10300; 10425	Aggregation: Count distinct for order counts; do not sum. Filter: Use exact numeric matching. Null: Not applicable; column is required. Join: Join to orderdetails.orderNumber .
orderDate date	Nullable: False Non-key attribute Role: date	Date the customer placed the order.	Synonyms: order date; purchase date; placed on; sales date Values: 2003-01-06; 2004-01-01; 2005-05-31	Aggregation: Group by year, quarter, month, or day. Filter: Use this date for sales-order timing and period-based revenue unless the user specifies shipment or payment timing. Null: Not applicable; column is required. Join: Not a join key.
requiredDate date	Nullable: False Non-key attribute Role: date	Requested or required delivery/shipment date for the order.	Synonyms: required date; due date; promised date; deadline Values: 2003-01-13; 2005-06-07	Aggregation: Group by period or use date differences. Filter: Compare with shippedDate to evaluate timeliness. Null: Not applicable; column is required. Join: Not a join key.
shippedDate date	Nullable: True Non-key attribute Role: date	Date the order was shipped.	Synonyms: ship date; shipped on; dispatch date Values: 2003-01-10; 2005-05-29	Aggregation: Group by shipment period or use date differences. Filter: Use NULL to identify orders not yet shipped in the snapshot. Null: Order has no recorded shipment date, usually because it is not shipped or is in a non-shipped status. Join: Not a join key.
status varchar(15)	Nullable: False Non-key attribute Role: categorical status	Current order status.	Synonyms: order status; state; fulfillment status Values: Shipped; In Process; Disputed; Cancelled; Resolved; On Hold	Aggregation: Group or count. Filter: Use exact standard values. Do not infer status from dates when status is available. Null: Not applicable; column is required. Join: Not a join key.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
comments text	Nullable: True Non-key attribute Role: unstructured text	Optional free-text notes about the order.	Synonyms: order comment; notes; remarks; issue description Values: Check on availability.; Difficult to negotiate with customer...	Aggregation: Do not aggregate. Filter: Use contains/full-text style matching for issue or keyword searches. Null: No order comment is recorded. Join: Not a join key.
customerNumber int(11)	Nullable: False Foreign key Role: relationship key	Customer who placed the order.	Synonyms: customer id; client id; buyer id; account number Values: 103; 141; 496	Aggregation: Group orders by customer; do not sum. Filter: Use exact customer number or join to customers for names/geography. Null: Not applicable; column is required. Join: Join to customers.customerNumber.

4.8 orderdetails

Business aliases: order detail; order line; line item; sales line; transaction line

Purpose: Stores the products, quantities, actual unit prices, and line positions for each order.

Row grain: One row per product line item within an order.

Standard row count: 2996

Primary key: orderNumber; productCode

Foreign keys: {'columns': ['orderNumber'], 'references': 'orders.orderNumber', 'optional': False}; {'columns': ['productCode'], 'references': 'products.productCode', 'optional': False}

Indexed columns: PRIMARY KEY (orderNumber, productCode); INDEX (productCode)

Common query intents: sales revenue; units sold; order value; product sales; line-item lookup; discount analysis; gross profit estimation

Common filters: orderNumber; productCode; quantityOrdered; priceEach; orderLineNumber

Common groupings: orderNumber; productCode; product line after join; customer after join; order period after join

Table Notes

- Line revenue equals quantityOrdered multiplied by priceEach.
- The composite primary key means the same product appears at most once per order in the standard schema.
- This table is the bridge between orders and products and creates a many-to-many business relationship.
- Always join to orders before filtering by order date or status.

Columns

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
orderNumber int(11)	Nullable: False Primary-key component and foreign key Role: identifier and relationship key	Order containing the line item.	Synonyms: order id; sales order number Values: 10100; 10250; 10425	Aggregation: Group line measures by order; do not sum the identifier. Filter: Use exact order number or join to orders. Null: Not applicable; column is required. Join: Join to orders.orderNumber.
productCode varchar(15)	Nullable: False Primary-key component and foreign key Role: identifier and relationship key	Product/SKU on the order line.	Synonyms: product id; SKU; item code Values: S18_1749; S24_3969; S50_1392	Aggregation: Group line measures by product; do not sum the identifier. Filter: Use exact text matching or join to products. Null: Not applicable; column is required. Join: Join to products.productCode.
quantityOrdered int(11)	Nullable: False Non-key attribute Role: additive quantity measure	Number of units ordered for this product on this order.	Synonyms: quantity; units ordered; units sold; sales volume Values: 18; 30; 50	Aggregation: Additive across line items when the selected order statuses and date scope are appropriate. Filter: Use numeric thresholds and aggregate by order, product, customer, date, or category as needed. Null: Not applicable; column is required. Join: Not a join key. Cautions: "Units sold" may need cancelled orders excluded depending on business intent.
priceEach decimal(10,2)	Nullable: False Non-key attribute Role: monetary unit measure	Actual unit selling price charged on the order line.	Synonyms: sale price; selling price; actual price; unit price; price charged Values: 35.29; 136.00; 214.30	Aggregation: Do not sum unit prices by themselves. Use weighted calculations or multiply by quantityOrdered for revenue. Filter: Use numeric comparisons; compare with products.MSRP for discounts and products.buyPrice for realized margin. Null: Not applicable; column is required. Join: Not a join key. Cautions: This is transactional sale price, unlike products.MSRP and products.buyPrice.

Column / Type	Null / Key	Business Meaning	Synonyms & Values	Aggregation / Filter / Join / Cautions
orderLineNumber smallint(6)	Nullable: False Non-key attribute Role: ordinal	Display or sequence position of the line within the order.	Synonyms: line number; item sequence; order position Values: 1; 3; 16	Aggregation: Do not sum. May order lines within one order. Filter: Filter only when the user explicitly asks for a specific line position. Null: Not applicable; column is required. Join: Not a join key. Cautions: Not globally unique and not a business measure.

5. Relationships and Cardinalities

ID	Parent → Child	Cardinality	Meaning	Join / Warning
R1	offices.officeCode → employees.officeCode	one office to many employees	Each employee works from one office; an office can employ multiple employees.	employees.officeCode = offices.officeCode Do not infer customer office directly; reach it through the assigned sales representative.
R2	employees.employeeNumber → employees.reportsTo	one manager to zero or many direct reports	Employees may report to another employee; the top-level employee has NULL reportsTo.	employee.reportsTo = manager.employeeNumber Use distinct aliases for employee and manager roles.
R3	employees.employeeNumber → customers.salesRepEmployeeNumber	one employee to zero or many assigned customers	A customer may be assigned to one sales representative; some customers are unassigned.	customers.salesRepEmployeeNumber = employees.employeeNumber Use an outer join when unassigned customers must remain visible.
R4	customers.customerNumber → orders.customerNumber	one customer to zero or many orders	Each order belongs to one customer; a customer may have no orders.	orders.customerNumber = customers.customerNumber Use an outer join to find customers without orders.
R5	customers.customerNumber → payments.customerNumber	one customer to zero or many payments	Each payment belongs to one customer; a customer may have no payments.	payments.customerNumber = customers.customerNumber Payments are customer-level and cannot be linked directly to an individual order.
R6	orders.orderNumber → orderdetails.orderNumber	one order to one or many line items	Each order line belongs to one order.	orderdetails.orderNumber = orders.orderNumber Joining orders to details changes the grain from order to line item; count distinct orders when needed.

ID	Parent → Child	Cardinality	Meaning	Join / Warning
R7	products.productCode → orderdetails.productCode	one product to zero or many order lines	Each order line references one product; products may have no recorded sales.	orderdetails.productCode = products.productCode Use an outer join from products to find unsold products.
R8	productlines.productLine → products.productLine	one product line to one or many products	Each product belongs to one product line.	products.productLine = productlines.productLine Product-line name is both identifier and display label.

6. Join Path Catalog

ID	Intent	Path	Keys / Critical Guidance	Result Grain
J1	Customer orders	customers → orders	customers.customerNumber = orders.customerNumber	One row per order before further aggregation.
J2	Customer order lines and revenue	customers → orders → orderdetails	customers.customerNumber = orders.customerNumber; orders.orderNumber = orderdetails.orderNumber	One row per order line.
J3	Customer products purchased	customers → orders → orderdetails → products	customers.customerNumber = orders.customerNumber; orders.orderNumber = orderdetails.orderNumber; orderdetails.productCode = products.productCode	One row per customer-order-product line.
J4	Product-line sales	orders → orderdetails → products → productlines	orders.orderNumber = orderdetails.orderNumber; orderdetails.productCode = products.productCode; products.productLine = productlines.productLine	One row per order line with product-line attributes.
J5	Sales-representative performance	employees → customers → orders → orderdetails	employees.employeeNumber = customers.salesRepEmployeeNumber; customers.customerNumber = orders.customerNumber; orders.orderNumber = orderdetails.orderNumber	One row per order line attributed to the customer's assigned representative.

ID	Intent	Path	Keys / Critical Guidance	Result Grain
J6	Office performance	offices → employees → customers → orders → orderdetails	offices.officeCode = employees.officeCode; employees.employeeNumber = customers.salesRepEmployeeNumber; customers.customerNumber = orders.customerNumber; orders.orderNumber = orderdetails.orderNumber	One row per order line attributed through sales representative to office.
J7	Customer payments	customers → payments	customers.customerNumber = payments.customerNumber	One row per payment.
J8	Customer ordered value versus paid value	customer_order_aggregate → customer_payment_aggregate	Aggregate orders/orderdetails by customer first; Aggregate payments by customer separately; Then join the two customer-level aggregates WARNING: Do not join raw orderdetails and raw payments together before aggregation; that produces many-to-many row multiplication and inflated totals.	One row per customer.
J9	Employee and manager	employees as employee → employees as manager	employee.reportsTo = manager.employeeNumber	One row per employee with optional manager.
J10	Unsold products	products → orderdetails	products.productCode = orderdetails.productCode using an outer join from products	One row per product before grouping or existence filtering.

7. Metric Catalog

ID / Metric	Meaning / Grain	Required Fields	Formula	Date & Status Policy	Cautions
M01 line_revenue	Extended sales value of one order line. Grain: order line	orderdetails.quantityOrdered; orderdetails.priceEach	quantityOrdered × priceEach	Date: orders.orderDate after joining orders Status: Include all statuses only when explicitly requested; for realized sales, normally exclude Cancelled and consider treatment of Disputed/On Hold.	Currency code is not stored.

ID / Metric	Meaning / Grain	Required Fields	Formula	Date & Status Policy	Cautions
M02 order_value	Total extended value of an order. Grain: order	orderdetails.orderNumber; quantityOrdered; priceEach	Sum of line_revenue for one orderNumber	Date: orders.orderDate Status: Apply the user's requested status scope.	No stored order total exists.
M03 customer_ordered_value	Total value ordered by a customer. Grain: customer	customers.customerNumber; orders.orderNumber; orderdetails.quantityOrdered; orderdetails.priceEach	Sum of line_revenue across the customer's orders	Date: orders.orderDate Status: Clarify whether cancelled/disputed/on-hold orders should count.	This is ordered value, not amount paid or recognized accounting revenue.
M04 units_ordered	Total units on selected order lines. Grain: selected grouping	orderdetails.quantityOrdered	Sum of quantityOrdered	Date: orders.orderDate Status: Usually exclude Cancelled when the phrase is units sold; include all when the phrase is units ordered.	Status is on orders, so join to orders before filtering.
M05 average_selling_price_weighted	Average actual unit selling price weighted by units. Grain: selected grouping	quantityOrdered; priceEach	$\text{Sum}(\text{quantityOrdered} \times \text{priceEach}) \div \text{Sum}(\text{quantityOrdered})$	Date: orders.orderDate Status: Use the same status scope in numerator and denominator.	A simple average of priceEach is not unit-weighted.
M06 gross_profit_estimate	Estimated profit using static product buyPrice as unit cost. Grain: selected grouping	quantityOrdered; priceEach; products.buyPrice	$\text{Sum}(\text{quantityOrdered} \times (\text{priceEach} - \text{buyPrice}))$	Date: orders.orderDate Status: Normally exclude Cancelled; make treatment of disputed/on-hold explicit.	buyPrice is not captured historically per order line, so this is an estimate.
M07 gross_margin_percent_estimate	Estimated gross profit as a percentage of sales value. Grain: selected grouping	line_revenue; gross_profit_estimate	$\text{gross_profit_estimate} \div \text{line_revenue} \times 100$	Date: orders.orderDate Status: Use the same status scope for revenue and profit.	Protect against zero revenue.
M08 discount_amount_vs_msrp	Difference between list-price value and actual line revenue. Grain: order line or grouping	quantityOrdered; products.MSRP; orderdetails.priceEach	$\text{quantityOrdered} \times (\text{MSRP} - \text{priceEach})$	Date: orders.orderDate Status: Apply the user's requested status scope.	Negative values indicate priceEach above MSRP.
M09 discount_percent_vs_msrp	Percent below MSRP for an order line. Grain: order line	products.MSRP; orderdetails.priceEach	$(\text{MSRP} - \text{priceEach}) \div \text{MSRP} \times 100$	Date: orders.orderDate Status: Status usually does not affect the unit calculation but affects included lines.	Protect against zero MSRP, although standard data uses positive values.
M10 inventory_units	Current units in stock across selected products. Grain: product or grouping	products.quantityInStock	Sum of quantityInStock	Date: Current snapshot; no historical date basis Status: Not applicable.	Do not trend over time because only one snapshot exists.

ID / Metric	Meaning / Grain	Required Fields	Formula	Date & Status Policy	Cautions
M11 inventory_cost_value	Estimated current inventory value at product buy price. Grain: product or grouping	quantityInStock; buyPrice	$\text{Sum}(\text{quantityInStock} \times \text{buyPrice})$	Date: Current snapshot Status: Not applicable.	Currency code is not stored.
M12 inventory_list_value	Estimated current inventory value at MSRP. Grain: product or grouping	quantityInStock; MSRP	$\text{Sum}(\text{quantityInStock} \times \text{MSRP})$	Date: Current snapshot Status: Not applicable.	This is not expected revenue and ignores discounts and sell-through.
M13 total_payments	Total amount received from customers. Grain: customer or time period	payments.amount	Sum of amount	Date: payments.paymentDate Status: Not applicable to order status.	Payments are not allocated to specific orders.
M14 average_payment_size	Average amount per payment transaction. Grain: customer or grouping	payments.amount; payments.checkNumber	Average amount across payment rows	Date: payments.paymentDate Status: Not applicable.	Different from average paid per customer.
M15 customer_balance_proxy	Difference between total ordered value and total payments for a matched scope. Grain: customer	customer_ordered_value; total_payments	$\text{customer_ordered_value} - \text{total_payments}$	Date: Use aligned date windows or lifetime scope on both sides Status: Order-side status policy must be explicit.	This is a proxy, not an accounting accounts-receivable balance.; Payment allocation and opening balances are absent.; Aggregate each side separately before joining.
M16 order_count	Number of distinct orders. Grain: selected grouping	orders.orderNumber	Count distinct orderNumber	Date: orders.orderDate unless another date is requested Status: Apply requested status scope.	After joining orderdetails, never use raw row count for order count.
M17 average_order_value	Average order total. Grain: selected grouping	order_value; orders.orderNumber	$\text{Sum of order values} \div \text{count of distinct orders}$	Date: orders.orderDate Status: Use a consistent status scope.	Compute order totals first or use mathematically equivalent grouped logic.
M18 fulfillment_days	Days from order placement to shipment. Grain: shipped order	orders.orderDate; orders.shippedDate	$\text{shippedDate} - \text{orderDate}$ in days	Date: orders.orderDate or shippedDate depending the user's wording Status: Only rows with a non-null shippedDate.	Cancelled or unresolved orders may have no shippedDate.
M19 days_early_or_late	Difference between actual shipment date and required date. Grain: shipped order	orders.requiredDate; orders.shippedDate	$\text{shippedDate} - \text{requiredDate}$ in days; positive is late, zero is on date, negative is early	Date: orders.shippedDate Status: Only rows with a non-null shippedDate.	Define the sign convention clearly in labels.

ID / Metric	Meaning / Grain	Required Fields	Formula	Date & Status Policy	Cautions
M20 on_time_shipment_rate	Share of shipped orders shipped on or before requiredDate. Grain: selected grouping	requiredDate; shippedDate; orderNumber	Count of shipped orders where $\text{shippedDate} \leq \text{requiredDate} \div \text{count of shipped orders}$	Date: orders.shippedDate Status: Exclude orders without shippedDate from the denominator unless the user defines a broader service-level metric.	Report denominator definition.
M21 customer_credit_utilization_proxy	Ordered-value proxy relative to credit limit. Grain: customer	customer_ordered_value; customers.creditLimit	$\text{customer_ordered_value} \div \text{creditLimit} \times 100$	Date: Aligned order period or lifetime Status: Order status scope must be explicit.	Not available credit and not accounting utilization.; Protect against zero or NULL creditLimit.
M22 product_sell_through_proxy	Historical units ordered relative to current stock snapshot. Grain: product	sum quantityOrdered; quantityInStock	$\text{units_ordered} \div (\text{units_ordered} + \text{current quantityInStock}) \times 100$	Date: Orders over selected historical period plus current stock snapshot Status: Usually exclude Cancelled for units sold.	Not true sell-through because historical opening inventory and replenishments are absent.

8. Query Intent Routing

Intent	Primary Tables	Optional Tables	Must Retrieve	Notes
Find customer details	customers	employees	customers table metadata	Join employees only when sales representative details are requested.
Find order details or status	orders	customers; orderdetails; products	orders metadata; status values	Use orderdetails only for products, quantities, or value.
Sales/revenue analysis	orders; orderdetails	customers; products; productlines; employees; offices	M01 line_revenue; status policy; orderDate semantics	Revenue is derived; it is not stored in orders.
Product performance	products; orderdetails; orders	productlines	products metadata; orderdetails metadata; status policy	Use orderDate/status from orders.
Inventory analysis	products	productlines	quantityInStock caveat	Current snapshot only.
Payment analysis	payments	customers	payments metadata; paymentDate semantics	Do not bring in orders unless comparing customer-level aggregates.
Customer balance or outstanding amount	customers; orders; orderdetails; payments		J8 separate aggregation rule; M15 customer_balance_proxy	This is a proxy, not a precise accounts-receivable balance.
Sales representative performance	employees; customers; orders; orderdetails	offices	J5; unassigned-customer caveat	Attribute orders through the customer's assigned sales representative.
Office or territory performance	offices; employees; customers; orders; orderdetails		J6; territory semantics	Office is inferred through sales representative assignment.

Intent	Primary Tables	Optional Tables	Must Retrieve	Notes
Employee hierarchy	employees		R2 self-reference	Use aliases for employee and manager roles.
Late or on-time shipments	orders	customers	M19; M20; shippedDate null semantics	Only shipped orders have a measurable actual shipment date.
Customers with no orders/payments	customers	orders; payments	outer-join guidance	Use existence/anti-join semantics; do not use inner joins.
Products never ordered	products; orderdetails	productlines	J10	Use products as the preserving side.
Discount analysis	orderdetails; products; orders		M08; M09	Compare actual priceEach with MSRP.
Profit or margin analysis	orderdetails; products; orders	customers; productlines	M06; M07; buyPrice historical-cost caveat	Results are estimates because buyPrice is static product metadata.

9. Ambiguity Resolution Rules

User Term	Default Interpretation	Clarify When
sales	Sum of quantityOrdered × priceEach using orders.orderDate.	The user might mean units, payments, shipped revenue, or only completed orders.
revenue	Derived line revenue, not payments.	The user needs recognized/accounting revenue or a specific status policy.
paid	payments.amount using paymentDate.	The user asks which specific order was paid.
order total	Sum of quantityOrdered × priceEach per order.	Taxes, shipping, or discounts outside line prices are expected; those fields do not exist.
price	Use priceEach for actual sold price, MSRP for list price, and buyPrice for unit cost.	The wording does not indicate which price concept is intended.
profit	Estimated gross profit using priceEach minus current buyPrice.	The user expects operating profit, tax, shipping, or historical cost.
customer balance	Customer ordered value minus payments as a proxy.	Precise accounts receivable is required.
top customer	Rank by ordered value.	The user may mean payments, order count, units, or credit limit.
best product	No universal default; choose only when the user states revenue, units, profit, order count, or margin.	Always when the metric is absent.
region	Customer geography when discussing customers; office territory when discussing organization or sales offices.	Both are plausible.
salesperson	employees joined through customers.salesRepEmployeeNumber.	The user means managers or all employees.
open orders	Orders with status In Process or On Hold and/or shippedDate NULL.	The user's organization uses a different status definition.
delayed order	shippedDate greater than requiredDate for shipped orders.	Unshipped past-due orders should also count.

User Term	Default Interpretation	Clarify When
inventory value	quantityInStock × buyPrice.	The user may want MSRP/list value.
year/month	Use orderDate for sales/order questions and paymentDate for payment questions.	Shipment timing is explicitly central.

10. Business Rules and Constraints

- Every order belongs to exactly one customer.
- Every order line belongs to exactly one order and one product.
- Every product belongs to exactly one product line.
- Every employee belongs to one office.
- A customer may have no assigned sales representative.
- An employee may have no manager only at the top of the hierarchy.
- A customer may have zero, one, or many orders and payments.
- A product may have zero or many order lines.
- orderdetails and payments use composite primary keys.
- Foreign-key delete/update cascade behavior is not explicitly declared; do not assume cascade operations.
- shippedDate and comments can be NULL.
- addressLine2, state, postalCode, salesRepEmployeeNumber, and creditLimit may be NULL in customers.
- addressLine2 and state may be NULL in offices.
- htmlDescription and image are normally NULL in the standard productlines data.
- order status values in the standard data are Shipped, In Process, Disputed, Cancelled, Resolved, and On Hold.
- productLine values in the standard data are Classic Cars, Vintage Cars, Motorcycles, Trucks and Buses, Planes, Ships, and Trains.
- office territory values are NA, EMEA, Japan, and APAC.
- orderLineNumber restarts within each order and is not globally unique.
- The same product cannot appear twice in the same order under the composite primary key.
- Monetary values contain two decimal places but no currency code.
- Dates are DATE values with no time-of-day or timezone.
- Text collation is case-insensitive in the source definition, but exact stored values should still be preferred for controlled categories.

11. SQL Generation Guardrails

- Generate read-only retrieval statements by default unless the user explicitly asks to modify data.
- Never invent a table, column, relationship, status, metric, or currency.
- Use only exact schema names and correct key relationships.
- Select the minimum necessary tables and columns.
- Avoid SELECT-all behavior unless the user explicitly requests every column.
- Before aggregation, identify the current row grain and the desired output grain.
- Use count distinct for order or customer counts after joining to lower-grain tables.
- Aggregate order data and payment data separately before comparing them at customer level.
- Use actual sale price from orderdetails.priceEach for sales calculations.
- Use products.MSRP only for list-price or discount comparisons.
- Use products.buyPrice only as a static cost estimate and label profit/margin results as estimated.
- Apply date filters to the date that matches the business question: orderDate, shippedDate, requiredDate, or paymentDate.

- Make order-status inclusion/exclusion explicit for revenue, units sold, profit, and order counts.
- Preserve rows with NULL relationships when the question asks for missing, unassigned, unsold, unpaid, or no-order entities.
- Treat text identifiers and postal/phone fields as strings.
- Do not link payments directly to orders.
- Do not claim historical inventory, returns, refunds, taxes, shipping charges, net income, or exact accounts receivable from this schema.
- Use clear business aliases in the result so derived values are understandable.
- If a request is ambiguous and materially changes the metric or scope, ask for clarification rather than guessing.
- When a safe default is possible, state the selected interpretation in the response metadata or explanation.

12. RAG Retrieval Design

Recommended strategy: Hybrid retrieval using semantic embeddings plus exact keyword/identifier matching. Retrieve table chunks, relationship chunks, and metric/rule chunks together.

Recommended top-k: Start with 5 to 8 chunks, then rerank for schema relevance and relationship coverage.

Chunking Principles

- Keep each table definition and its columns in one coherent chunk or a table-summary chunk plus column-detail chunk.
- Keep relationship and join-path guidance separate from individual table chunks so multi-table questions retrieve it.
- Keep metric definitions in domain groups such as sales, payments, inventory, and fulfillment.
- Keep ambiguity and guardrail rules in a dedicated policy chunk.
- Include exact table names, column names, aliases, and user synonyms in every relevant chunk.
- Attach neighboring chunk identifiers so the retriever can expand context when a query spans multiple entities.

Recommended Chunks

Chunk ID	Title	Tags	Neighbors
CM-00-OVERVIEW	Database overview and global limitations	classicmodels; overview; tables; limitations; currency; date range	CM-01-RELATIONSHIPS; CM-10-GUARDRAILS
CM-01-RELATIONSHIPS	Foreign keys, cardinalities, and join paths	joins; foreign keys; relationships; grain; cardinality	CM-00-OVERVIEW; CM-02-CUSTOMERS; CM-05-ORDERS; CM-06-ORDERDETAILS
CM-02-CUSTOMERS	Customers table metadata	customers; clients; accounts; credit limit; sales rep; geography	CM-03-EMPLOYEES; CM-05-ORDERS; CM-07-PAYMENTS
CM-03-EMPLOYEES	Employees and hierarchy metadata	employees; sales reps; manager; reportsTo; jobTitle	CM-04-OFFICES; CM-02-CUSTOMERS
CM-04-OFFICES	Offices and territory metadata	offices; branches; territory; country; city	CM-03-EMPLOYEES; CM-09-METRICS-ORG
CM-05-ORDERS	Orders table and status/date semantics	orders; status; orderDate; requiredDate; shippedDate; comments	CM-06-ORDERDETAILS; CM-02-CUSTOMERS; CM-08-METRICS-SALES
CM-06-ORDERDETAILS	Order line metadata and revenue grain	orderdetails; line items; quantityOrdered; priceEach; revenue	CM-05-ORDERS; CM-11-PRODUCTS; CM-08-METRICS-SALES
CM-07-PAYMENTS	Payments table and payment limitations	payments; amount; checkNumber; paymentDate; paid	CM-02-CUSTOMERS; CM-09-METRICS-PAYMENTS

Chunk ID	Title	Tags	Neighbors
CM-08-METRICS-SALES	Sales, revenue, discount, and margin metrics	sales; revenue; order value; profit; margin; discount; average order value	CM-05-ORDERS; CM-06-ORDERDETAILS; CM-11-PRODUCTS
CM-09-METRICS-PAYMENTS-ORG	Payment, balance, employee, office, and fulfillment metrics	payments; balance; sales rep performance; office performance; on time; late	CM-07-PAYMENTS; CM-03-EMPLOYEES; CM-04-OFFICES; CM-05-ORDERS
CM-10-GUARDRAILS	Ambiguity rules and SQL-generation guardrails	ambiguity; guardrails; no hallucination; status policy; date basis; grain	CM-00-OVERVIEW; CM-01-RELATIONSHIPS
CM-11-PRODUCTS	Products table metadata	products; SKU; inventory; MSRP; buyPrice; vendor; scale	CM-12-PRODUCTLINES; CM-06-ORDERDETAILS; CM-08-METRICS-SALES
CM-12-PRODUCTLINES	Product lines and category descriptions	productlines; category; Classic Cars; Motorcycles; Planes; Ships; Trains	CM-11-PRODUCTS

Retrieval Query Expansion Examples

User Phrase	Expand To
top clients by sales	customers; orders; orderdetails; customer_ordered_value; line_revenue; status policy
unpaid customers	customers; orders; orderdetails; payments; customer_balance_proxy; separate aggregation
best salesman	employees; customers.salesRepEmployeeNumber; orders; orderdetails; performance metric ambiguity
late deliveries	orders.requiredDate; orders.shippedDate; days_early_or_late; unshipped past due ambiguity
low stock classic cars	products.quantityInStock; products.productLine; Classic Cars; current inventory snapshot

13. Data Quality and Limitations

Issue	Impact
Static sample dataset	Results are for learning/demo use and not current operational data.
No currency code	Monetary results must not be labeled with a currency unless configured externally.
No payment allocation to orders	Exact paid/unpaid status per order cannot be determined.
No historical cost	Gross profit using buyPrice is an estimate based on current/static product cost.
No inventory history	Inventory trends, stock at order date, and replenishment analysis are impossible.
Nullable geography fields	NULL state/postal values may be normal for international addresses, not necessarily errors.
Inconsistent text cleanliness	Some values may contain trailing spaces, punctuation variants, or spelling quirks; trim for display and cautious matching.

Issue	Impact
Free-text comments/descriptions	Keyword searches may miss synonyms and can produce false positives; semantic retrieval can help.
Status interpretation	Ordered, sold, shipped, and realized revenue are different concepts; status policy must be explicit.
Attribution through assigned rep	Office and employee performance reflects customer assignment, not necessarily the person who processed each individual order.
Product vendor denormalized	Vendor details beyond the vendor name cannot be queried.
Legacy utf8 definition	The source uses MySQL utf8/utf8_general_ci; application-level ingestion should preserve text consistently.

14. Business Glossary

Term	Definition
Actual selling price	orderdetails.priceEach, the unit price charged on a specific order line.
Buy price	products.buyPrice, the product-level unit acquisition cost used for estimated margin.
MSRP	products.MSRP, the suggested/list price, not actual revenue.
Line revenue	quantityOrdered multiplied by priceEach.
Order value	Sum of line revenue across all lines of one order.
Customer ordered value	Sum of order values for a customer within the selected scope.
Payment total	Sum of payments.amount within the selected paymentDate scope.
Customer balance proxy	Ordered value minus payment total; not a formal accounting balance.
Sales representative	Employee referenced by customers.salesRepEmployeeNumber.
Office performance	Sales attributed to office through employee assignment and customer assignment.
Product line	One of seven product categories stored in productlines and products.
Open order	A business interpretation typically involving In Process or On Hold status and/or no shippedDate; definition should be confirmed.
On-time shipment	A shipped order with shippedDate on or before requiredDate.
Unassigned customer	A customer with NULL salesRepEmployeeNumber.
Unsold product	A product with no matching rows in orderdetails.
Current inventory	products.quantityInStock as a single present-state snapshot in the dataset.

15. Source References

MySQLTutorial ClassicModels overview: <https://www.mysqltutorial.org/getting-started-with-mysql/mysql-sample-database/>

Used for: Database purpose and official list of eight tables.

ClassicModels SQL dump, version 3.2: <https://gist.github.com/prof3ssorSt3v3/796ebc82fd8eeb0b697effaa1e86c3a6>

Used for: Exact table definitions, column types, indexes, foreign keys, representative values, and source version notes.

ClassicModels practice reference: <https://www.scribd.com/document/527411619/CECS323ClassicModelsPracticeSQL>

Used for: Standard row counts for all eight tables.

ClassicModels database description: <https://www.richardtwatson.com/open/Reader/ClassicModels.pdf>

Used for: Business interpretation of product lines and pricing fields.

End of canonical ClassicModels metadata. The companion JSON and TXT files contain the same metadata content.